

AI FitTrack Project Testing

Phase 6

This test report consolidates the supplied automated, smoke, Atlas, Postman and live Gemini evidence while separating recorded results from items that still need final visual capture.

Field	Value
Project	AI FitTrack
Submission	SmartBridge Naan Mudhalvan
Project scope	Backend-only REST API
Technology	Node.js, Express.js, MongoDB, Mongoose, JWT and Google Gemini AI
Source baseline	Latest uploaded backend ZIP and supplied verification evidence
Document status	Submission-ready content; student metadata and final video links remain pending

Prepared for academic evaluation. Secrets, credentials, private environment files and unverified claims are excluded.

Document contents

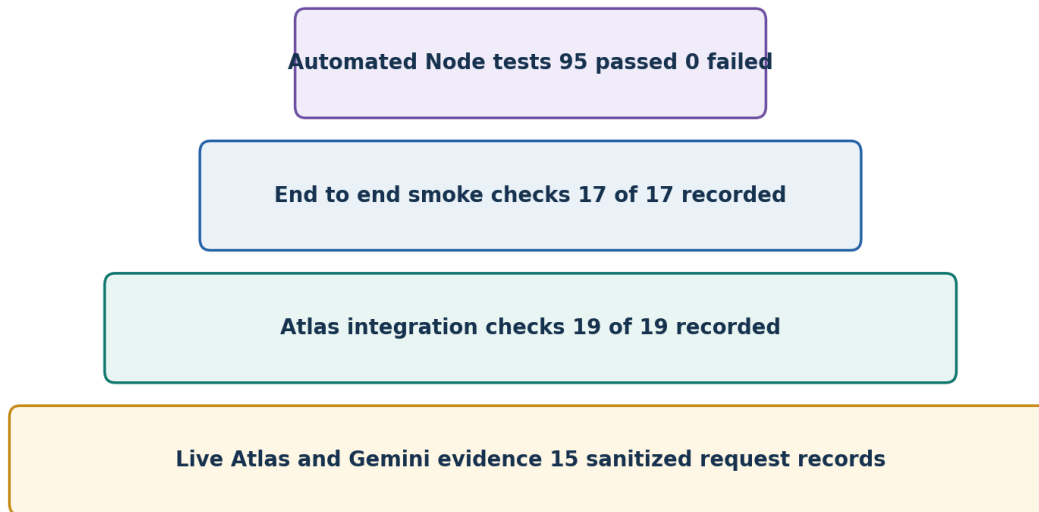
1. Test strategy
2. Automated coverage
3. Live verification
4. Detailed cases
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Test strategy

Testing is layered so that pure validation and service behavior are checked quickly, HTTP routes are tested against isolated MongoDB instances, smoke and Atlas scripts verify real integration, and sanitized live calls prove the main end-to-end flows including Gemini.

Verification layers and recorded results

Counts are taken from supplied evidence; no new result is invented



Coverage includes auth validation security isolation CRUD search AI retries and error handling

Figure 1 Verification layers and result counts recorded in the supplied evidence

Recorded result summary

Verification layer	Recorded result	Evidence status
Automated Node tests	95 passed, 0 failed	Recorded in verification files; 95 test definitions are present in six test files.

Verification layer	Recorded result	Evidence status
End-to-end smoke checks	17 of 17	Recorded in README verification.
Atlas checks	19 of 19	Recorded in README and requirements verification.
Live authentication	4 sanitized records	201, 200, 200 and 401 results supplied.
Live workouts	8 sanitized records plus negative follow-ups	CRUD and three search filters supplied.
Live Gemini	3 sanitized records	Recommendation, submitted insights and database insights supplied.
Postman collection	12 requests	Health, three auth, six workout and two AI requests.

Automated test distribution

Test file	Cases	Coverage
ai.test.js	23	AI validation, structured responses, database stats, failures, rate limiting and regression
auth.test.js	23	Registration, login, profile, token failures, duplicate and password behavior
geminiRetry.test.js	13	Transient retries, Retry-After, bounded backoff, non-retryable failures and parsing
security.test.js	8	CORS, ownership, secret handling, password exposure and security boundaries
validation.test.js	16	Authentication, workout and AI input normalization and validation
workouts.test.js	12	CRUD, search, ordering, isolation, validation and UTC date behavior

Test file	Cases	Coverage
Total	95	Recorded automated run: 95 passed and 0 failed

Live API evidence register

Area	Case	Endpoint	Status	Verified outcome
Authentication	Register	POST /api/auth/register	201	Pass - token returned and password omitted
Authentication	Login	POST /api/auth/login	200	Pass - same account returned
Authentication	Authorized profile	GET /api/auth/profile	200	Pass - profile returned
Authentication	Missing token	GET /api/auth/profile	401	Pass - no user data returned
Workouts	Create	POST /api/workouts	201	Pass - owner matched authenticated user
Workouts	List	GET /api/workouts	200	Pass - count and record matched
Workouts	Get by ID	GET /api/workouts/:id	200	Pass
Workouts	Update	PUT /api/workouts/:id	200	Pass - changed values persisted
Workouts	Search name	GET /api/workouts/search?name=	200	Pass - count 1
Workouts	Search category	GET /api/workouts/search?category=	200	Pass - count 1
Workouts	Search date	GET /api/workouts/search?date=	200	Pass - UTC day match
Workouts	Delete	DELETE /api/workouts/:id	200	Pass - follow-up GET returned 404
Gemini	Recommendation	POST /api/ai/workout-recommendation	200	Pass - 7-day structured response

Area	Case	Endpoint	Status	Verified outcome
Gemini	Submitted insights	POST /api/ai/fitness-insights	200	Pass - supplied metrics referenced
Gemini	Database insights	POST /api/ai/fitness-insights	200	Pass - exact 3, 38.3 and 780 calculation

Detailed test cases

ID	Case	Input or condition	Expected	Recorded result
T01	Valid registration	Unique name, email and password	201 and safe user plus JWT	Pass recorded
T02	Duplicate registration	Existing email	409 conflict	Pass recorded
T03	Valid login	Correct email and password	200 and JWT	Pass recorded
T04	Invalid login	Wrong or unknown credentials	Generic 401	Pass recorded
T05	Protected profile	Valid Bearer token	200 and safe profile	Pass recorded
T06	Profile without token	No Authorization header	401	Pass live
T07	Create workout	Valid workout fields	201 and JWT owner	Pass live
T08	Invalid workout	Missing or invalid fields	400 field errors	Pass recorded
T09	List own workouts	Authenticated request	200 newest first	Pass live
T10	Cross-user isolation	User B requests user A data	No foreign record	Pass recorded
T11	Search by name	Partial case-insensitive term	Only matching own records	Pass live
T12	Search by category	Running	Exact canonical category	Pass live
T13	Search by date	Valid calendar date	UTC-day match	Pass live
T14	Get by ID	Owned ObjectId	200 correct workout	Pass live
T15	Invalid ID	Malformed identifier	400	Pass recorded
T16	Update workout	Duration and calories	200 changed values	Pass live

ID	Case	Input or condition	Expected	Recorded result
T17	Delete workout	Owned ObjectId	200 then 404	Pass live
T18	Recommendation	Age, goal and experience	Structured Gemini plan	Pass live
T19	Recommendation validation	Unsupported age or goal	400	Pass recorded
T20	Submitted insights	Workout summary metrics	Structured analysis	Pass live
T21	Database insights	useDatabaseStats true	Exact owner-only aggregate	Pass live
T22	No workouts insights	Empty owner history	200 with insights null	Pass recorded
T23	AI burst limit	Requests over configured limit	429 and Retry-After	Pass recorded
T24	Gemini transient failure	Retryable service error	Bounded retry	Pass recorded
T25	Gemini bad key	Non-retryable auth error	Single attempt and safe error	Pass recorded
T26	Malformed AI output	Invalid JSON from provider	Safe 502	Pass recorded
T27	CORS denied origin	Origin outside CLIENT_URL	403	Pass recorded
T28	Unknown route	Unregistered API path	404 standard envelope	Pass recorded

Sanitized evidence excerpts

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Registration
POST /api/auth/register -> HTTP 201
success: true
message: Registration successful
verified: token present; password and hash absent

Unauthorized profile
GET /api/auth/profile without token -> HTTP 401
success: false
message: Authentication required: missing Authorization header

Workout flow
POST /api/workouts -> 201 and owner match true
GET /api/workouts -> 200 count 1
PUT /api/workouts/:id -> 200 duration 45 calories 400
Search name/category/date -> 200 count 1 for each
DELETE /api/workouts/:id -> 200
Follow-up GET -> 404

Gemini database insights
POST /api/ai/fitness-insights { "useDatabaseStats": true } -> 200
totalWorkouts: 3

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averageWorkoutDuration: 38.3
 totalCaloriesBurned: 780
 verified calculation: $(30 + 45 + 40) / 3 = 38.3$; $280 + 320 + 180 = 780$

Observations and evidence limitations

Observation	Impact	Required action
No screenshot images or raw terminal logs are included in the evidence ZIP.	The text evidence is detailed, but some portals expect visual proof.	Capture fresh Postman, terminal test and Atlas screenshots without revealing secrets.
Delete evidence shows data:null; uploaded controller returns data.workout.	Core delete result remains verified; exact response shape is not synchronized.	Re-run delete on the frozen source and replace the evidence capture.
Submitted insights evidence uses statsSource manual; uploaded source uses submitted.	AI computation is verified; one label differs.	Re-run the submitted-statistics request and capture the current label.
No deployment URL is supplied.	A local demo is possible; hosted verification is not documented.	Add a public base URL only if the course requires deployment.

Final re-test checklist

- Freeze the GitHub commit and install dependencies from package-lock.json.
- Run npm test and capture the final 95-test summary.
- Run npm run test:smoke and npm run test:atlas with temporary test data.
- Run the Postman collection from health through Gemini in order.
- Re-capture delete and submitted-statistics responses from the exact frozen commit.
- Remove temporary users and workouts, then confirm cleanup counts.
- Review every image for hidden tokens, keys, passwords, connection strings and personal data before submission.

Items requiring student confirmation

Pending item	What to provide
Student identity	Name, NM ID, register or roll number and department were not present in the supplied project files.
Institution details	College name, academic year, guide or mentor name and official submission date need confirmation.

Pending item	What to provide
Team details	Team name, member list and responsibility allocation were not supplied; no ownership is assigned in these documents.
Repository	Insert the final public GitHub repository URL after upload.
Deployment	Insert the deployed API base URL if the evaluator requires a hosted demonstration.
Demonstration video	Record the video using Phase 8 and insert the final shareable video URL.